
Unsupervised Wavetable Seam Artifact Repair via Hybrid GA–PPO Meta-Search

A PREPRINT

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19 July 2026

Code (ReelSynth engine + DenoiseOpt meta-search): <https://github.com/reeldemo/reelsynth>,
<https://github.com/reeldemo/denoise-opt-meta>

ABSTRACT

When a single-period wavetable is played with phase wrap, a sample discontinuity at the period boundary injects a broadband transient that repeats at the fundamental. We repair this cycle-local seam artifact by baking seam-local operators into the period inside a **bake cell**: classical DualCosine crossfade, polish, FIR blends, and optional tiny residual networks, optionally mixed by MoE soft gates. Candidates are ranked by the prolonged residual score $R \in [0, 1]$ (1 = best), the clamped tiled RMS agreement between the baked multi-period waveform and a procedural ideal sibling that withholds the open-wrap cliff. **DenoiseOpt** is the residual-scored hybrid meta-search: GA tournament updates, PPO architecture proposals, optional PBT exploit–mutate, and MoE routing over bake cells. Studio-clean pairs are not required for every trial. On the frozen canonical sine+cliff holdout (seed 20,260,719), a compact favorite reaches $R \approx 0.991$ versus DualCosine $R \approx 0.825$ ($\Delta R \approx +0.166$). Separately, a live GPU campaign at the 5,000-iteration clean gate reports champion $R \approx 0.9909$ against DualCosine ≈ 0.8166 on the runner geometry. Scope is cycle-local wavetable seam restoration.

Keywords unsupervised learning · wavetable · wrap discontinuity · seam restoration · DenoiseOpt · residual score · neural architecture search · reinforcement learning · genetic algorithms · meta-learning

1 Introduction

Wavetable oscillators store one period of a waveform and advance a read pointer modulo the table length. If $x[0] \neq x[L-1]$, the wrap is a hard discontinuity. Under cyclic playback it becomes a periodic click whose spectrum is broadband and locked to the fundamental. The same geometry appears whenever a short loop buffer is tiled for listening or for objective evaluation. We call the repair task *cycle-local seam restoration*: edit a neighborhood of the wrap (and optionally a lightweight residual cell over the period) so that multi-period playback matches an ideal reference that never opens the wrap cliff. That outer agreement is scored by the prolonged residual R for this periodic seam artifact class.

Figure 1 shows one real sine-wrap edge case from the frozen canonical holdout (same `make_batch` tile, seed 20,260,719). Panel (a) is the ideal multi-period sine. Panel (b) is the cracked engine tile after open-wrap cliff modulation ($R \approx 0.950$), with the ideal overlaid so the seam jump is visible. Panel (c) applies DualCosine on that same tile ($R \approx 0.603$ on this severe cliff). Panel (d) is the top DenoiseOpt favorite (champion_iter_000235, batch $R \approx 0.991$) on the same tile ($R \approx 0.9917$). Panels (e)–(f) plot the modulation resid-

ual and a seam zoom so the favorite’s correction is readable against DualCosine.

Classical virtual-analog practice already supplies seam-local tools. Alias-free BLIT/BLEP-style synthesis [1], Poly-BLEP / fractional-delay antialiasing [3], and BLAMP corner corrections [2] reduce wrap and geometric discontinuity energy in-band. Those operators act locally. They leave open which bake parameters and which residual cell, if any, minimize prolonged tiled error against a no-cliff ideal. Label-free restoration work such as Noise2Noise shows that useful reconstructors can be ranked without paired clean targets when the evaluation geometry is explicit [4]. Speech-domain Noise2Noise denoisers extend that idea to audio waveforms [5]. Mixture-invariant unsupervised separation likewise avoids paired clean stems for ranking [29]. On the search side, NAS surveys [16], RL controllers [17, 18], aging evolution [20, 21], population-based hyperparameter schedules [22], CMA-ES tutorials [24], and evolution-guided policy gradients [23] motivate a hybrid outer loop: GA for population diversity, PPO for discrete architecture edits [19], and PBT-style exploit–mutate on elites. Platform-aware and progressive NAS add compute-aware search priors [27, 28]. MoE soft gates suggest routing among heterogeneous tiny cells un-

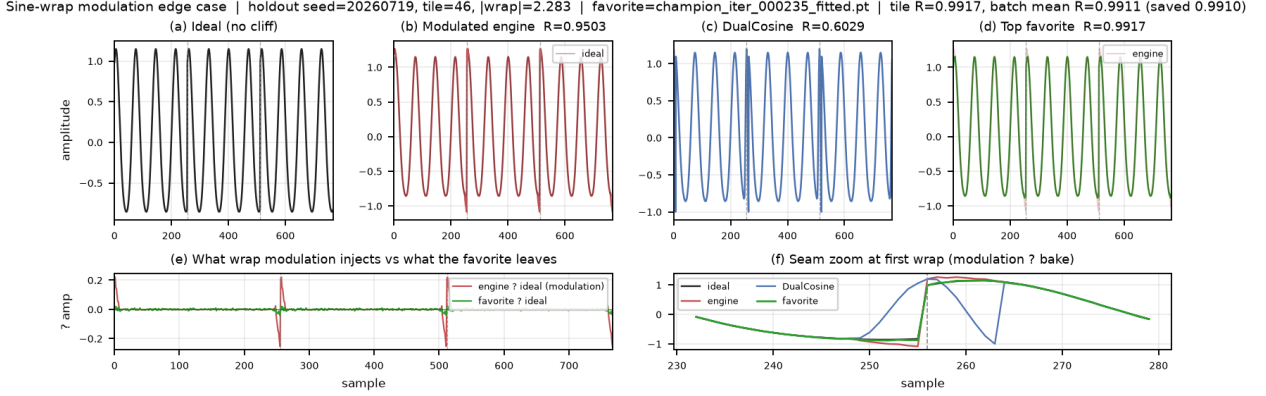


Figure 1: Canonical holdout sine-wrap edge case (seed 20,260,719, $L=256$, $N=16$ tiles, same tile across panels). (a) Ideal sibling (cliff withheld). (b) Engine with open-wrap cliff (ideal overlay). (c) DualCosine bake. (d) DenoiseOpt favorite bake. (e)–(f) Modulation residual and seam zoom. Prolonged residual $R \in [0, 1]$ ($1 = \text{best}$) is printed per panel where scored.

der one residual score [25]. Waveform separation nets (Wave-U-Net, Conv-TasNet, dual-path RNN, SepFormer) and audio DL surveys inform the searchable cell family at bake-scale width [7–9, 11, 12, 30, 31], with U-Net / ResNet / attention primitives as block ancestors [10, 13–15].

The contribution is a meta-learning protocol for this periodic seam instance. An outer RL+GA(+PBT) loop proposes bake-cell operator graphs and continuous bake parameters. Each candidate is scored by prolonged residual $R \in [0, 1]$ ($1 = \text{best}$) between engine-baked multi-period audio and a procedural ideal tiling. Paired studio-clean wavetables are not required for every trial. Differentiable DSP modules [6] place DualCosine / FIR / MLP seam ops in an interpretable stack. Bayesian HPO [26] informs prior families that compete under the same R . A CUDA overnight campaign (PPO+GA+PBT+NAS, depth and MoE biases) has passed a 5,000-iteration clean gate with champion $R \approx 0.9909$ versus DualCosine ≈ 0.8166 on the runner geometry. The frozen canonical holdout and inference benches in Results are the measured evaluation matrices for this report. We do not claim general speech or music enhancement SOTA.

Contributions.

1. **Problem framing.** Cast wavetable wrap discontinuity as cycle-local seam restoration under prolonged tiled evaluation, separating seam-local diagnostics from residual R .
2. **Hybrid DenoiseOpt outer loop.** Specify RL+GA(+PBT) search over discrete operator graphs and continuous bake parameters, with named branches (GA crossover, PPO mutation, PBT exploit, MoE soft gates).
3. **Formal residual protocol.** Define ideal-tile siblings, R with range/perfect-match/monotone properties, and reimplementable Algorithms 1–8.

4. **Open evaluation path.** Release companion code for the synthesizer engine and meta search, with frozen 5k-gate figures and matched classical-vs-favorite benches.

Section 2 organizes related work by mechanism. Sections 3–4 detail the residual objective and search protocol. Results report the 5k-gate freeze and inference benches.

2 Related Work

We organize prior work by mechanism. **Used** lines are methodological or comparative anchors for residual-scored RL+GA meta-search. **Screened** lines were inspected as contrast. They are excluded from the dependency set. Paraphrases prefer attached abstracts and PDF snippets over title-only citation.

2.1 Discontinuity control and modular audio DSP

Alias-free digital synthesis of classic analog waveforms (BLIT/BLEP lineage) states the wrap discontinuity problem for trivial oscillators [1]. PolyBLEP-style fractional-delay corrections give practical antialiasing oscillators for subtractive synthesis [3]. BLAMP corner rounding treats discontinuities in the first derivative (triangular edges, clipping, rectification) without heavy oversampling [2]. DDSP reframes DSP modules as composable blocks inside learning pipelines [6]. We use that framing for small bake/FIR/MLP seam cells. We do not train end-to-end neural vocoders for every meta trial. These works are **used** as domain anchors for the operator vocabulary and for why prolonged cyclic playback is the outer judge.

2.2 Label-free restoration

Noise2Noise shows that restoration mappings can be learned from corrupted observations alone by exploiting structure in the corruption process [4]. Speech Noise2Noise denoisers apply the same idea to waveforms without clean paired audio [5]. Mixture invariant training ranks unsupervised separation systems without stem labels [29]. That philosophy

motivates ranking denoise candidates without paired studio-clean wavetable cycles. The residual compares an engine-baked tiling to a procedural ideal that withholds open-wrap cliffs. We **use** this family as conceptual justification for unsupervised residual ranking. We do not retrain a Noise2Noise CNN on images.

2.3 Waveform denoising and separation cells

Deep learning for audio surveys the move from hand-designed features to learned time/frequency models [7]. Wave-U-Net performs end-to-end source separation with multi-scale 1-D U-Nets [8]. Improved Wave-U-Net speech enhancement and singing-voice U-Nets popularized encoder-decoder skips on audio [9, 10]. The original U-Net, ResNet residuals, and Transformer attention supply the block primitives we shrink to bake width [13–15]. Conv-TasNet and dual-path RNN demonstrated strong time-domain separation with temporal convolutions and long-context paths [11, 12]. SepFormer and audio spectrogram transformers push attention-heavy audio stacks further [30, 31]. These lines are **used** as design priors for tiny searchable cells (shallow U-Net / dilated / dual-path / attention blocks under a strict per-trial bake budget). They are not frozen full-scale separators inside every meta trial.

2.4 NAS and RL controllers

Elsken et al. taxonomize NAS by search space, strategy, and evaluation cost [16]. Their performance-estimation axis includes lower-fidelity proxies such as shorter training, data subsets, and lower-resolution images [16]. Zoph and Le couple RL controllers to architecture search [17]. ENAS adds parameter sharing for efficient discrete search [18]. Progressive and platform-aware NAS add staged evaluation and latency-aware rewards [27, 28]. We **use** this family as the ancestor of RL proposing architecture edits (add/remove/mutate ops, toggle residual-primary, mutate MLP/FIR), scored with prolonged residual. PPO supplies the clipped-surrogate policy update when the overnight loop logs PPO mutation branches [19].

2.5 Evolutionary NAS and population schedules

Large-scale and regularized / aging evolution showed that evolutionary NAS can discover competitive image classifiers when population turnover is carefully regularized [20, 21]. Population Based Training jointly adapts weights and hyperparameters in an asynchronous population [22]. Practical Bayesian optimization [26] and CMA-ES tutorials [24] inform local Bayesian and continuous evolutionary prior families. These are **used** for GA tournament/crossover branches, PBT-style exploit-mutate, and literature-combo priors, implemented at synthesizer-trial scale.

2.6 Evolutionary reinforcement learning hybrids

Khadka and Tumer’s evolution-guided policy gradient (ERL) interleaves an evolutionary population with an RL actor to mitigate sparse reward and brittle hyperparameters in deep RL [23]. We **use** ERL as the spirit of interleaving GA pop-

ulation steps with PPO updates at tiny population sizes. We do not claim a full ERL robotics stack.

2.7 Mixture-of-experts soft gating

Sparsely gated MoE layers increase capacity via conditional computation [25]. We **use** MoE as motivation for soft gates over heterogeneous seam blocks (e.g., U-Net vs dilated vs attention cells) under residual scoring, at bake-scale widths.

2.8 Used versus screened

Used (methodological or comparative anchors).

- Discontinuity / VA lineage: [1–3].
- Unsupervised restoration philosophy: [4, 5, 29].
- Modular / differentiable audio DSP context: [6].
- Audio DL and compact waveform cells: [7–15, 30, 31].
- NAS / RL controllers / PPO: [16–19, 27, 28].
- Evolutionary NAS and population HPO: [20–22, 24, 26].
- ERL hybrid outer loop: [23].
- MoE soft gating prior: [25].

Screened (contrast only).

- **MAML** [32]: bi-level / fast-adaptation framing. We do not differentiate through a deep task network. Inner fits are coordinate-style updates on bake parameters. Outer rank is residual R .
- **DARTS** [33]: continuous relaxation of architecture search. We keep discrete ops + small cells under residual scoring.
- **Demucs / Hybrid Demucs / realtime waveform enhancement** [34–36]: strong separators and enhancers, but full multi-scale / hybrid STFT stacks are too heavy per meta trial under overnight bake budgets.
- **DiffWave** [37]: diffusion vocoder. Screened as a generative sampling stack outside the seam bake operator set.
- **SEGAN-style adversarial enhancement** [38]: optional tiny adversarial cues only. Full GAN training loops screened for per-trial cost.

Positioning. Relative to this map, the present work contributes (i) a prolonged residual objective for periodic seam denoising, (ii) an explicitly hybrid RL+GA(+PBT) meta-outer loop with honest branch naming, and (iii) a used/screened literature protocol with downloaded source PDFs. Claims stay inside cycle-local seam restoration under a declared compute budget. The primary claim is protocol-level: residual-scored meta-search for cycle-local audio denoise operators.

3 Methods

3.1 Notation

Throughout, $x \in \mathbb{R}^L$ is the stored single-period wavetable (raw engine input). $y = \Theta(x; \theta)$ is the engine-baked cycle after seam ops and optional residual cells with parameters θ . $r^* \in \mathbb{R}^L$ is a procedural ideal sibling from the same seed and family with the open-wrap cliff withheld. Studio-clean recordings are not used as r^* . Tiling length N (typically $N=16$) forms $y_{\text{tiled}} = \text{Tile}(y, N)$ and $r^*_{\text{tiled}} = \text{Tile}(r^*, N)$. Seam width W (code: SEAM_W, typically 8) is the neighborhood edited by classical fade and polish ops. $R \in [0, 1]$ is the prolonged residual score defined below (1 = best). These symbols are fixed for the residual objective. They follow a different convention from Noise2Noise clean/noisy pairs.

3.2 Ideal tile construction

The ideal sibling is generated by the same procedural draw as the engine input (frequency, phase, mid-cycle noise policy) with the open-wrap cliff amplitude set to zero at both ends. Engine and ideal therefore share the mid-cycle content of one seed draw. They differ by the seam cliff (and any seam-boosted noise policy applied only to the engine path). This sibling relationship is what makes unsupervised ranking possible without paired studio recordings [4, 5].

3.3 Residual score R

With tiling length N ,

$$R = \text{clamp}\left(1 - \frac{\text{rms}(y_{\text{tiled}} - r^*_{\text{tiled}})}{\max(\text{rms}(r^*_{\text{tiled}}), \varepsilon)}, 0, 1\right), \quad (1)$$

where $\text{rms}(z) = \sqrt{\frac{1}{|z|} \sum_i z_i^2}$ and $\varepsilon > 0$ avoids division by zero. Optional soft shape gates (when enabled) multiply R by factors that down-weight mid-cycle damage relative to seam-local error. The propositions below concern the ungated score in (1).

Proposition 1 (Range). *For any finite y, r^* and $\varepsilon > 0$, the score in (1) satisfies $R \in [0, 1]$.*

Proof. The ratio of nonnegative RMS terms is nonnegative, so $1 - \text{rms}(\cdot)/\max(\cdot, \varepsilon) \leq 1$. The outer $\text{clamp}(\cdot, 0, 1)$ forces the lower bound 0. $\square$

Proposition 2 (Perfect match). *If $y_{\text{tiled}} = r^*_{\text{tiled}}$, then the pre-clamp identity equals 1, hence $R = 1$.*

Proof. $\text{rms}(0) = 0$, so $1 - 0/\max(\text{rms}(r^*_{\text{tiled}}), \varepsilon) = 1$, and clamp leaves 1. $\square$

Proposition 3 (Monotone RMS). *Fix r^*_{tiled} . If $\text{rms}(y_{\text{tiled}} - r^*_{\text{tiled}}) < \text{rms}(y'_{\text{tiled}} - r^*_{\text{tiled}})$ and both pre-clamp scores lie in $(0, 1)$, then the pre-clamp score of y is strictly larger than that of y' .*

Proof. The map $e \mapsto 1 - e/d$ with fixed $d = \max(\text{rms}(r^*_{\text{tiled}}), \varepsilon) > 0$ is strictly decreasing in the error RMS e . $\square$

Algorithm 1 ResidualScore

Require: Ideal cycle r^* , baked cycle y , periods N

- 1: $I \leftarrow \text{Tile}(r^*, N)$
 - 2: $Y \leftarrow \text{Tile}(y, N)$
 - 3: $r_{\text{rms}} \leftarrow \text{RMS}(Y - I)$
 - 4: $i_{\text{rms}} \leftarrow \max(\text{RMS}(I), \varepsilon)$
 - 5: **return** $\text{clamp}(1 - r_{\text{rms}}/i_{\text{rms}}, 0, 1)$
-

Algorithm 2 DualCosine baseline

Require: Engine cycle x

- 1: Fade both ends of x with raised-cosine windows of width W
 - 2: **return** $\text{ResidualScore}(r^*, x_{\text{faded}}, N)$
-

Algorithm 3 MoE soft gating over bake experts

Require: Cycle x , expert cells $\{E_k\}_{k=1}^K$, gate logits $g \in \mathbb{R}^K$

- 1: $w \leftarrow \text{softmax}(g)$
 - 2: $y \leftarrow \sum_{k=1}^K w_k E_k(x)$
 - 3: **return** y
-

Lemma 1 (Wrap closure, sufficient sketch). *Suppose the only difference between engine and ideal is an endpoint cliff, and a bake zeros the endpoint jump while changing mid-cycle samples by at most ε_{mid} in RMS. Then R is nondecreasing relative to the untreated engine whenever the mid-cycle penalty is dominated by the seam error reduction under (1).*

Sketch. This is a conditional accounting argument under the stated assumptions. It does not assert that every DualCosine-style fade improves R on every draw. Empirical support appears in the classical and favorite benches of Section 5. $\square$

Remark 1 (Noise2Noise motivation). Ranking without studio-clean pairs is possible when the ideal is a procedural sibling of the engine draw [4, 5]. The remark is a design philosophy for procedural siblings.

3.4 Bake operator and DualCosine baseline

A **bake cell** $\Theta(\cdot; \theta)$ composes classical seam ops (DualCosine, polish, soft seam, FIR blends) with optional tiny residual cells (MLP/FIR/U-Net-lite/attention-lite) and optional MoE soft gates over experts. Bake parameters θ live in a bounded box. Architecture strings are discrete graphs over those ops.

3.5 Search space

Each genome encodes: ordered op list (fade, polish, FIR, pin, residual write), depth (number of composed blocks), residual cell type (none, MLP, FIR, U-Net-lite, attn-lite), optional MoE expert count $K \in \{2, 3, 4\}$, and continuous θ in box bounds used by the overnight runner (fade widths, FIR taps, residual scales). The live tag is PPO+GA+PBT+NAS+depth+MoE.

3.6 Training / fit inner loop

3.7 Hybrid outer loop

Each iteration rotates among branches: PPO, GA, PBT, NAS, and combo. Selection is always by residual R (plus logged

Algorithm 4 FitCell (inner loop)

Require: Architecture cfg, fit steps T , batch size B

```

1: cell  $\leftarrow$  SeamCell(cfg); opt  $\leftarrow$  Adam(cell.params)
2: prev  $\leftarrow$  null; patience  $\leftarrow$  0
3: for  $t = 1, \dots, T$  do
4:   ideal, eng  $\leftarrow$  MakeBatch( $B$ )
5:   out  $\leftarrow$  ApplyOps(eng, cell, cfg.ops)
6:    $R \leftarrow$  meanResidualScore(ideal, out)
7:   loss  $\leftarrow$   $1 - R$ 
8:   if finite(loss) then AdamStep(opt, loss)
9:   end if
10:  if prev  $\neq$  null and  $|\text{prev} - R| / \max(|\text{prev}|, \epsilon) < 10^{-4}$  then
11:    patience  $\leftarrow$  patience + 1; if patience  $\geq 3$  then
12:      break
13:    else
14:      patience  $\leftarrow$  0
15:    end if
16:  end for
17: return cell,  $R$ 

```

Algorithm 5 GA tournament step

Require: Population P , tournament size k , crossover rate p_c , mutation rate p_m

```

1: Sample  $k$  genomes; parents  $\leftarrow$  top-2 by  $R$ 
2: child  $\leftarrow$  Crossover(parents) with prob.  $p_c$ , else clone elite
3: Mutate(child) with rate  $p_m$  (ops, depth, cell type,  $\theta$ )
4: Fit and evaluate child; insert into  $P$  (replace worst if full)
5: return updated  $P$ 

```

Algorithm 6 PPO architecture update

Require: Actor-critic π_ϕ , clip ϵ , experience buffer of mutate actions

```

1: Sample action  $a$  editing ops / hyperparameters from  $\pi_\phi$ 
2: Fit proposed cfg; observe reward  $\rho = R - R_{\text{DualCosine}}$ 
3: Advantage  $\hat{A}$  from critic; ratio  $r_\phi = \pi_\phi(a) / \pi_{\phi_{\text{old}}}(a)$ 
4: Update  $\phi$  with clipped surrogate  $\min(r_\phi \hat{A}, \text{clip}(r_\phi, 1 - \epsilon, 1 + \epsilon) \hat{A})$ 
5: return updated  $\pi_\phi$ 

```

auxiliaries). Reward for PPO is R minus the DualCosine baseline on the same batch.

Remark 2 (Trial complexity). Per outer iteration the dominant cost is $O(T_{\text{fit}} \cdot C_{\text{fwd}})$ for fit steps times one forward bake. Bake-width caps (tiny residual cells) keep C_{fwd} far below full-band Demucs/DiffWave forwards screened in Related Work.

3.8 Hyperparameters

Table 1 lists the overnight CUDA defaults used for the 5k-gate freeze reported in Results. Canonical companion pseudocode lives in docs/PSEUDOCODE.md (mirrored Algorithms 1–8).

Algorithm 7 PBT exploit–mutate

Require: Population P , exploit quantile q , mutate scale σ

```

1: Rank  $P$  by  $R$ ; for each bottom member, copy hyperparameters from a top- $q$  elite
2: Perturb copied hyperparameters by multiplicative noise of scale  $\sigma$ 
3: Refit briefly; write back into  $P$ 
4: return updated  $P$ 

```

Algorithm 8 HybridMetaSearch

Require: Iteration budget T_{out} , population size n

```

1:  $P \leftarrow$  InitPopulation( $n$ );  $\pi \leftarrow$  ActorCritic(); champ  $\leftarrow$  null; boredom  $\leftarrow$  0
2: for  $it = 1, \dots, T_{\text{out}}$  do
3:   branch  $\leftarrow$  Rotate(ppo, ga, pbt, nas, combo)
4:   cfg, hp  $\leftarrow$  Propose(branch,  $P$ ,  $\pi$ )
5:   cell,  $R_{\text{fit}} \leftarrow$  FitCell(cfg, hp.fit_steps)
6:    $R_{\text{eval}} \leftarrow$  EvalCell(cell)
7:    $R \leftarrow \frac{1}{2}R_{\text{fit}} + \frac{1}{2}R_{\text{eval}}$  (+ optional depth/MoE bonus)
8:   UpdatePopulation( $P$ , cfg,  $R$ ); PPOUpdate( $\pi$ , reward =  $R - R_{\text{DualCosine}}$ )
9:   if  $R >$  champ. $R$  then champ  $\leftarrow$  (cfg, cell,  $R$ ); boredom  $\leftarrow$  0
10:  else boredom  $\leftarrow$  boredom + 1
11:  end if
12:  if boredom  $\geq$  1000 then PlateauAdapt; boredom  $\leftarrow$  0
13:  end if
14: end for
15: return champ

```

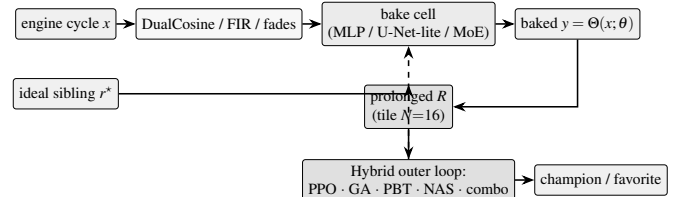


Figure 2: DenoiseOpt bake path and residual-scored hybrid meta-search. Classical seam ops and optional residual cells form a bake cell. Prolonged R ranks proposals. The outer loop rotates PPO/GA/PBT/NAS/combo branches. Dashed arrow: architecture proposals update the cell.

4 Experiments

Protocol freeze. Evaluation follows paper/v5/EVAL_PROTOCOL.md (v1). Primary metric: prolonged R . Secondary: SNR, SDR, and wrap-jump on tiled audio. Holdout seed 20,260,719. Search seed 1,902,771,841. We report the 5k-gate freeze plus frozen holdout. We make no unfinished-budget claims. PESQ/STOI stay out of scope on non-speech cycles.

4.1 Dataset

Creation. The paper-facing evaluation corpus is a frozen draw from the same synthetic generator used

Table 1: Overnight hybrid search hyperparameters (RTX 3090 runner defaults).

Parameter	Value
Population size n	1000
GA tournament k	10
Crossover / mutation rates p_c, p_m	0.7, 0.3
PPO clip ϵ	0.2
Adam learning rate (fit)	1e-3
Entropy bonus (PPO)	small positive
PBT exploit quantile / mutate scale	top 1/3, $\sigma \approx 0.2$
MoE experts K	$\{2, 3, 4\}$ when enabled
Plateau boredom threshold	1000 iters
Cycle length L / tiles N / seam W	256 / 16 / 8
Algo tag	PPO+GA+PBT+NAS+depth+MoE

by the overnight CUDA runner (`make_batch` in `overnight_gpu_rl_arch.py`). Each cycle has length $L=256$. A two-harmonic sine is drawn with frequency $\sim \mathcal{U}(1, 4)$ and phase $\sim \mathcal{U}(0, 2\pi)$. An open-wrap seam cliff of amplitude $\pm \mathcal{U}(0.08, 0.43)$ is applied over `SEAM_W=8` samples at both ends. Seam-boosted Gaussian noise (scale 0.02 at the ends, $0.15\times$ in the mid-cycle) is added to form the engine input. The ideal withholds the cliff and is used only for residual scoring. Prolonged residual R tiles each cycle $N=16$ times before the RMS ratio. The holdout seed is 20,260,719 (distinct from the overnight search seed 1,902,771,841). Overnight search draws fresh i.i.d. batches from this generator and does not train on the frozen holdout tensors. There is no supervised clean/noisy training split of studio audio.

Multi-family diversity (≥ 20 waveforms). Beyond the single sine+cliff holdout, Phase 3a scores methods on 20 generative draws: ten family variants (`sine_cliff`, `harmonic_fft`, `am_fm`, `nonlinear`, `combo`, `triple_mix`, `extreme_overlay`, `open_wrap_bias`, `soft_noise`, `wide_seam`) $\times$ two seeds (20,260,719 and 20,260,720), batch 64 each. These Python generative variants stress harmonic / AM-FM / nonlinear / overlay / open-wrap behavior of `make_batch` and remain the primary SOTA matrix. Separately, `export_sound_bench_tiles` exports 20 Rust `sound_bench` engine/ideal pairs (10 families $\times$ 2 seeds, $L=256$, byte-aligned with `generate_sound` / `generate_sound_ideal`) scored under the same residual geometry as a secondary transfer block (Table 7). Every method sees the same tensors per waveform. DualCosine is the paired baseline for ΔR and Wilcoxon tests.

Dataset metrics. A metrics draw of 4,096 cycles under the holdout seed yields mean ideal RMS 0.721 ± 0.016 , mean engine residual RMS 0.021 ± 0.013 , and mean absolute wrap jump $|x_0 - x_{L-1}|$ of 0.913 ± 0.634 . Identity (no bake) on that draw scores mean $R \approx 0.971$. Method tables use a matched score batch of 64 cycles from the same seed (tensors under `brand/artifacts/canonical_eval_dataset/`),

five consecutive seeds for mean $\pm$ std on the classical table, and the 20-waveform matrix for SOTA secondary metrics. Figure 1 is the documented sine-wrap edge case: a high-value wrap holdout tile where cliff modulation is visible, scored with DualCosine and the top neural favorite under the same residual R (tile $R \approx 0.9917$, favorite batch mean $R \approx 0.991$).

Overnight hybrid campaign (5k clean gate). A CUDA search tagged PPO+GA+PBT+NAS+depth+MoE runs with a large declared iteration budget (order 10^5). At the reported freeze (run `gpu_rl_arch-20260719T083019Z`, $\approx 6,922$ clean iters / ≈ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ against DualCosine baseline ≈ 0.81658 ($\Delta R \approx +0.174$). Overnight branch-best residuals at the freeze: GA ≈ 0.99016 , PBT ≈ 0.99012 , PPO ≈ 0.98924 , combo ≈ 0.98854 , NAS ≈ 0.98677 . Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full, 150 iters, same search seed, plateau adapt off) are reported beside those freezes in Table 5. Figures in Section 5 regenerate from `history.jsonl` near this gate.

Inference, classical, and learned baselines. High- R fitted cells ($R \geq 0.98$) are timed on CUDA (batch 64, prolonged residual). Ten non-learnable bake denoisers share the frozen canonical holdout against the neural favorite (Table 3). Fixed-arch MLP-on- R and CNN/U-Net-lite-on- R baselines (no outer NAS) are trained on $1-R$ with the same generator and scored in the SOTA matrix. Family-conditional hardness uses the separate Rust procedural 100,000-cycle bench reported in Results.

5 Results

Family-conditional hardness (completed bench). On the fitted DenoiseOpt 100,000-cycle bench, denoise score D is lowest for nonlinear ($D \approx 0.39$) and combo ($D \approx 0.40$), then `triple_mix` / `extreme_overlay` ($D \approx 0.51$ – 0.52), while `harmonic_fft` and `am_fm` sit near $D \approx 0.69$ – 0.70 . Lit-combo champion family stress (prolonged residual) similarly ranks `open_wrap_bias` ($R \approx 0.83$) and `extreme_overlay` / `combo` ($R \approx 0.88$ – 0.89) below `harmonic_fft` ($R \approx 0.95$). Hard sounds recur.

Overnight hybrid (5k-gate freeze). At the reported overnight freeze ($\approx 6,922$ clean iterations, ≈ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ with DualCosine ≈ 0.81658 on the overnight runner ($\Delta R \approx +0.174$). Monitoring plots below are regenerated from that freeze. Declared larger budgets continue. This paper reports the gate freeze only.

Inference time at matched score. Across diverse high- R fitted cells ($R \geq 0.98$), we timed CUDA forward passes (batch 64, prolonged residual). Within a tight band of the best residual ($R \geq R_{\max} - 5 \cdot 10^{-4}$), the **favorite** is the lowest-latency model: `champion_iter_000235` with $R \approx 0.9910$, ≈ 3.15 ms/batch, ≈ 37 k parameters (versus a higher-capacity max- R cell at $R \approx 0.9914$ but ≈ 7.18 ms/batch / ≈ 104 k

Table 2: Top-5 distinct fitted architectures on the CUDA inference bench (batch 64, prolonged R , seed geometry matching overnight). Favorite is latency-best inside the near-max- R band ($R \geq R_{\max} - 5 \cdot 10^{-4}$).

Tag	R_{saved}	R_{live}	ms/batch	params
iter_000157	0.9914	0.9913	7.18	104 484
iter_000205	0.9911	0.9907	7.29	70 562
iter_000235*	0.9910	0.9915	3.15	37 489
iter_000397	0.9910	0.9907	6.80	97 262
iter_000229	0.9909	0.9902	6.67	100 500

*Favorite (fastest in $R \geq R_{\max} - 5 \cdot 10^{-4}$ band).

Table 3: Method scores on the canonical sine+cliff holdout (batch 64, seed 20,260,719, $L=256$, $N=16$). Mean $\pm$ std over five consecutive seeds. Identity is a no-op ceiling. Primary metric is prolonged R (1 = best).

Method	R	R (5-seed)	ms/batch
neural favorite	0.9911	0.9912 ± 0.0002	2.56
identity (no-op)	0.9718	0.9726 ± 0.0021	0.31
seam_fir3	0.9610	0.9616 ± 0.0019	0.67
classic quadratic	0.8449	0.8453 ± 0.0117	1.59
DualCosine	0.8249	0.8258 ± 0.0137	1.35
cosine fade	0.8249	0.8258 ± 0.0137	1.59
crossfade	0.8116	0.8130 ± 0.0145	1.50
hann blend	0.8048	0.8060 ± 0.0156	0.54
soft wide fade	0.7432	0.7378 ± 0.0171	2.64
detrend	0.4079	0.4091 ± 0.0386	0.36
ensemble detrend+DC+FIR	0.4024	0.4027 ± 0.0387	1.56

params). Selection rule: maximize R , then minimize latency inside the band.

5.1 Top-5 architectures

Table 2 ranks five distinct fitted bake graphs from the CUDA inference bench (`inference_bench.json`, batch 64), preferring residual R first and latency on ties, and skipping near-duplicate copies of the same topology signature.

Classical vs favorite (canonical holdout). Table 3 scores the classical bake set plus DualCosine and the neural favorite on the frozen corpus of Section 4.1 (CUDA, batch 64, seed 20,260,719). DualCosine reaches $R \approx 0.825$ (≈ 1.35 ms/batch), near the overnight monitor DualCosine ≈ 0.817 on live draws from the same generator. The best *active* classical method is seam-local 3-tap FIR (`seam_fir3`) at $R \approx 0.961 / \approx 0.67$ ms/batch (0.962 ± 0.002 over five seeds). Identity (no-op) scores $R \approx 0.972$ and is a ceiling only. The neural favorite reaches $R \approx 0.991$ (0.991 ± 0.0002 over five seeds) at ≈ 2.56 ms/batch (≈ 37 k params): $\Delta R \approx +0.030$ vs `seam_fir3` and $\Delta R \approx +0.166$ vs DualCosine.

5.2 SOTA matrix (multi-family + secondary metrics)

Table 4 reports mean $\pm$ std over the 20-waveform multi-family set (Section 4.1), with canonical-holdout SNR/SDR/wrap-

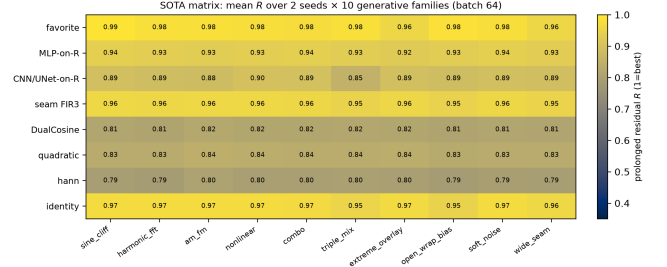


Figure 3: Colorblind-safe (cividis) heatmap of mean prolonged R for selected methods $\times$ generative families (2 seeds each). Higher is better.

jump for the same method set from `sota_matrix.json`. PESQ/STOI/MUSHRA are omitted for non-speech cycles (Limitations). Fixed MLP-on- R and CNN/U-Net-lite-on- R (no outer NAS) sit between DualCosine and the meta favorite. Paired Wilcoxon signed-rank (normal approx.) of favorite vs DualCosine on the 20 waveforms: $W=0$, $p \approx 8.9 \times 10^{-5}$, mean $\Delta R = +0.162$ (bootstrap 95% CI [0.155, 0.170]).

5.3 Ablations and compute

Table 5 reports two views. Overnight *branch-best freezes* at the 5k gate come from one hybrid history. Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full) use 150 iters on the same search seed with plateau adapt off. Short-budget champions sit below the multi-hour freeze by construction. They are labeled as such. Table 6 summarizes the overnight compute budget. Peak VRAM is from an `nvidia-smi` poll during a short full-branch search replay (30 iters) plus a champion forward+FitCell replay. Overnight `history.jsonl` did not log memory.

5.4 Rust sound_bench transfer (20 tiles)

Table 7 scores the same classical set and the Python-trained neural favorite on 20 Rust-exported tiles (10 families $\times$ 2 seeds). This closes the prior gap where multi-family tables used only Python generative stand-ins. Honest transfer note: the favorite was searched on Python `make_batch` sine+cliff geometry. On Rust families it still beats DualCosine (mean $\Delta R + 0.063$, Wilcoxon $p \approx 0.009$) but trails identity and `seam_fir3`. The primary SOTA claim remains the Python multi-family matrix.

6 Discussion

Where residual mass remains. Across the 100,000-cycle procedural bench and lit-combo family stress tests, wrap error is not uniform. The same generator families repeatedly dominate the residual budget: `nonlinear`, `combo`, `triple_mix`, and `extreme_overlay` show the weakest denoise / residual scores, while `harmonic_fft` and `am_fm` are comparatively easy. `open_wrap_bias` can look strong on wrap-energy proxies yet still lag on prolonged residual. Cliff reduction and ideal-matched tiling are related objectives with different rankings.

Table 4: SOTA comparison on 20 generative waveforms (10 families $\times$ 2 seeds, batch 64). R /SNR/SDR/wrap-jump are multi-family mean $\pm$ std. Columns marked * use the canonical freeze (seed 20,260,719) for ms/batch and ΔR vs DualCosine. Primary metric R (1 = best). No PESQ/STOI/MUSHRA.

Method	R (20-wave)	SNR (dB)	SDR (dB)	wrap-jump	ms*	params	ΔR^*
neural favorite	0.977 ± 0.010	35.0 ± 3.2	35.1 ± 3.2	0.90 ± 0.14	2.44	37 489	+0.166
identity	0.965 ± 0.008	30.9 ± 1.9	31.0 ± 1.9	0.91 ± 0.15	0.00	0	+0.147
seam_fir3	0.955 ± 0.005	28.1 ± 1.2	28.1 ± 1.2	0.46 ± 0.08	0.59	0	+0.136
MLP-on- R	0.932 ± 0.006	24.6 ± 0.9	24.6 ± 0.9	0.64 ± 0.07	2.76	8 604	+0.113
CNN/U-Net-on- R	0.886 ± 0.012	19.5 ± 0.9	19.5 ± 0.9	0.48 ± 0.06	5.01	11 752	+0.069
classic quadratic	0.835 ± 0.012	17.9 ± 1.1	17.7 ± 1.2	0.32 ± 0.05	1.78	0	+0.020
DualCosine	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	1.48	0	0
cosine fade	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	2.64	0	0
crossfade	0.802 ± 0.013	16.7 ± 1.2	16.4 ± 1.2	0.91 ± 0.15	1.70	0	-0.013
hann blend	0.793 ± 0.015	16.1 ± 1.2	15.8 ± 1.2	0.32 ± 0.05	0.57	0	-0.020
soft wide fade	0.729 ± 0.016	14.0 ± 1.1	13.6 ± 1.1	0.63 ± 0.08	3.69	0	-0.082
detrend	0.375 ± 0.040	5.5 ± 1.0	5.4 ± 1.0	0.00 ± 0.00	0.12	0	-0.417
ensemble DC+FIR	0.368 ± 0.040	5.1 ± 1.0	5.0 ± 1.0	0.12 ± 0.02	1.99	0	-0.422

Table 5: Ablations. Left block: overnight branch-best freeze (run gpu-rl-arch-20260719T083019Z, final iter $\approx$ 6,922). Right block: isolated short-budget re-runs (150 iters, seed 1,902,771,841, plateau adapt off). R is prolonged residual on the runner geometry.

Config	Overnight freeze R	Isolated 150-iter R
Full hybrid	0.99093	0.98413
GA-only	0.99016	0.98602
PBT (branch best / n/a isolated)	0.99012	0.98602
PPO-only	0.98924	0.97922
GA+PPO	—	0.98607
combo (branch best)	0.98854	0.98607
NAS (branch best)	0.98677	0.98607
DualCosine (runner)	0.81658	0.81658

Table 6: Compute budget for the reported overnight freeze (RTX 3090, PyTorch 2.6+cu124, search seed 1,902,771,841). Arch evaluations $\approx$ clean iterations.

Item	Value
GPU	NVIDIA GeForce RTX 3090
Elapsed wall time	$\approx$ 7.92 h
Clean iterations / arch evals	$\approx$ 6,922
Population size	12
Final champion R	0.99093
DualCosine baseline (runner)	0.81658
Peak VRAM (nvidia-smi replay)	$\approx$ 3.29 GiB

— Peak memory: max `nvidia-smi memory.used` during a 30-iter full-branch search probe ($\approx$ 3.29 GiB) and champion forward+FitCell replay ($\approx$ 3.17 GiB). The multi-hour overnight process itself left no memory log. Idle GPU occupancy was already $\approx$ 3.0 GiB from other sessions.

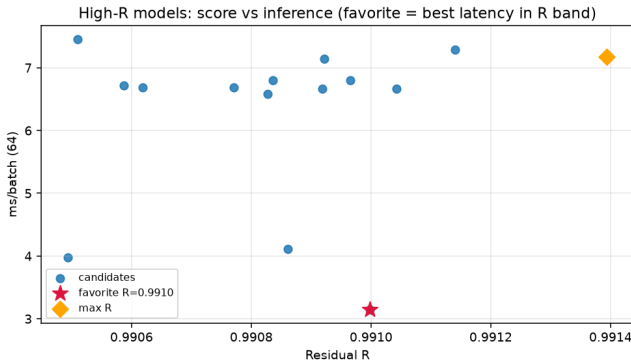


Figure 4: High- R fitted cells on CUDA (batch 64): prolonged residual R versus forward latency (ms/batch). Star marks the favorite (fastest inside the near-max- R band).

Implication for this paper. Mean R (and mean DualCosine gaps) therefore compress a structured difficulty map. Reporting only the overnight mean on synthetic sine cliffs under-

Table 7: Secondary transfer on 20 Rust `sound_bench` tiles (byte-aligned export). Mean $\pm$ std prolonged R . Favorite vs DualCosine: mean ΔR +0.063, $p \approx 0.009$.

Method	R (20 Rust tiles)	ΔR vs DualCosine
identity	0.928 ± 0.084	+0.166
seam_fir3	0.896 ± 0.077	+0.134
neural favorite	0.825 ± 0.121	+0.063
classic quadratic	0.788 ± 0.110	+0.026
DualCosine / cosine fade	0.762 ± 0.118	0

states how much of the remaining $\sim 1\%$ sits in a few hard families. That structure is a first-class empirical claim of the present work.

Hybrid search vs. plateau. Overnight traces at the 5k gate show early high champions then long plateaus near $R \approx 0.99$. Plateau adaptation (deeper cells, denser MoE graphs, higher GA/PBT weight) is the operational response inside one cam-

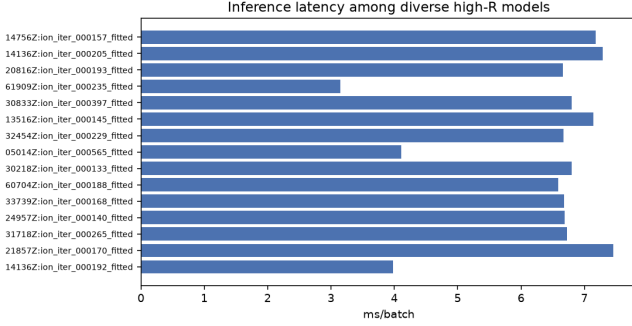


Figure 5: Inference latency (ms/batch, CUDA, batch 64) for diverse high- R fitted models from the inference bench.

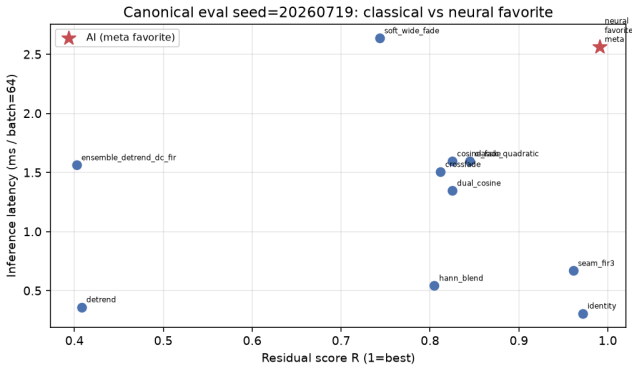


Figure 6: Canonical holdout (seed 20,260,719, batch 64): classical bake methods and neural favorite in the R vs CUDA ms/batch plane. Star: favorite.

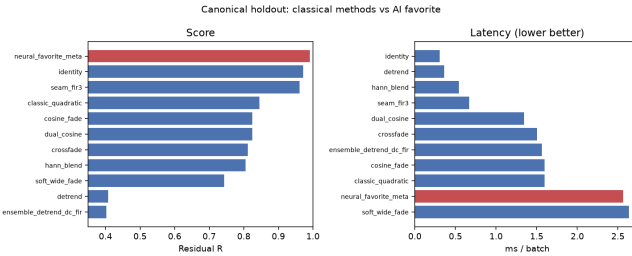


Figure 7: Canonical holdout ranking on prolonged residual R (left) and CUDA latency (right). DualCosine $R \approx 0.825$. Favorite highlighted.

paign. It does not by itself rebalance family-conditional risk. A natural next paper is family-conditional and cliff-conditional meta-learning: train or select denoisers per family, or condition the outer loop on measured wrap statistics, and publish per-family residual curves beside the scalar champion.

Hybrid branches. At the reported overnight freeze, GA and PBT lead branch-best residuals, with PPO close behind and NAS trailing on this seed (Table 5). Isolated short-budget re-runs on the same seed give measured champions under a fixed 150-iter cap (Table 5). Overnight branch tags are freezes

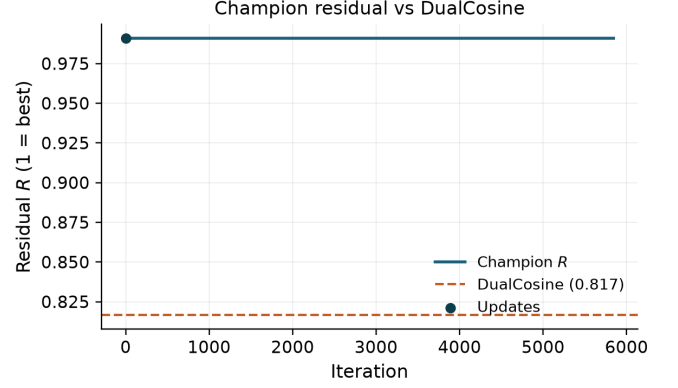


Figure 8: Overnight 5k-gate freeze: champion residual R versus clean search iteration, with DualCosine baseline on the runner geometry (RTX 3090).

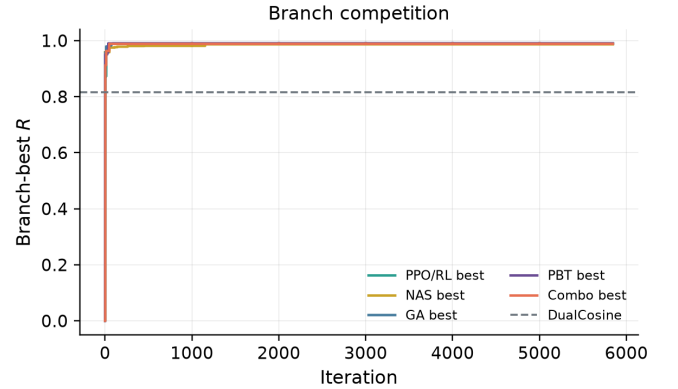


Figure 9: Overnight 5k-gate freeze: per-branch best residual trajectories (PPO/RL, GA, PBT, NAS, combo) versus iteration.

from one hybrid run. They are not causal single-branch experiments. The multi-family SOTA matrix (Table 4) shows the meta favorite ahead of DualCosine on R , SNR, and SDR across twenty Python generative draws. Fixed MLP/CNN-on- R baselines beat DualCosine and trail the searched favorite. On Rust `sound_bench` tiles (Table 7) the same favorite beats DualCosine and trails identity. That transfer gap matches the narrow Python training geometry.

7 Tooling and AI use

This section discloses research and writing tools, including AI systems, with scope and intent.

Literature discovery. Open literature was gathered with the Klaut Research Gateway (local OpenAlex / Crossref / arXiv providers) under queries spanning virtual-analog anti-aliasing, unsupervised restoration, HPO/PBT, RL-GA hybrids for NAS, and audio denoise architectures. Harvest JSONs live in the companion meta repository. Source PDFs were downloaded for every bibliography en-

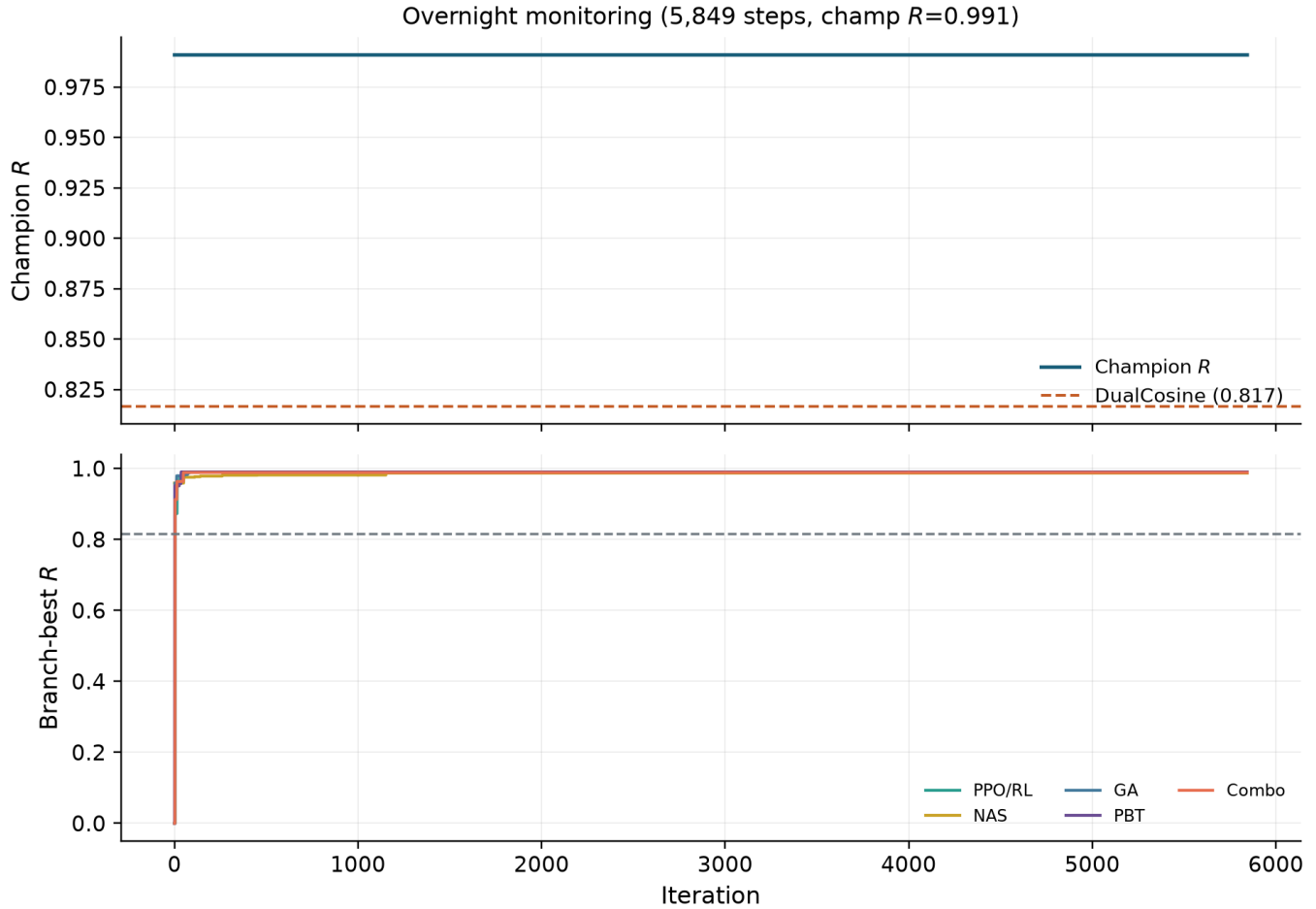


Figure 10: Full-width overnight monitoring panel at the 5k-gate freeze: champion R (top) and branch-best trajectories (bottom). Source: `history.jsonl` regenerated figures.

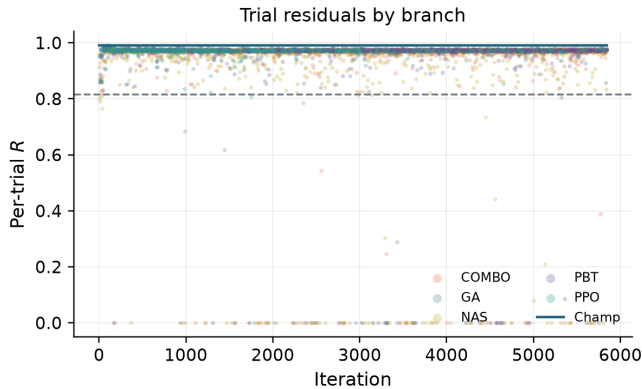


Figure 11: Per-trial residual scatter by search branch at the 5k-gate freeze, with champion overlay.

try (`artifacts/literature_oa/pdfs/`). Paywalled-only items were dropped or replaced.

Paper drafting. Section planning, subsection drafts, and revision passes used Klaut `research_paper_*` tools with

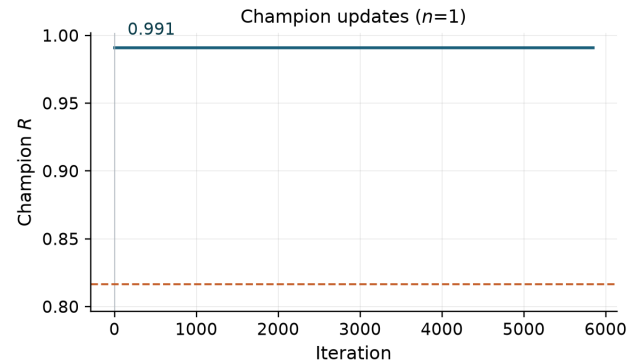


Figure 12: Champion update timeline at the 5k-gate freeze. Sparse R labels mark first, mid-peak, and latest improvements for twocolumn legibility.

local Ollama (`qwen3.5:9b`) when available, under an explicit scientific workflow (plan $\rightarrow$ write $\rightarrow$ revise $\rightarrow$ export) into an arXiv-style twocolumn template. AI drafts were treated as scaffolding. Claims, numbers, and citations were

checked against harvest records, PDF analyses, and measured artifacts. Final prose was edited for DSP terminology and to remove generic filler.

Why AI was used. (1) Scale literature triage across DSP and AutoML without pretending every high-cite hit is in-domain. (2) Accelerate IMRaD scaffolding so human effort focuses on metric honesty (residual vs DualCosine, family hardness, gate vs final overnight claims). (3) Keep an auditable tool chain (plans, PDF snippet analyses, citation flags) over opaque chat-only writing.

Where AI was *not* used. Model fitting, residual scoring, DualCosine baselines, lit-combo / overnight GPU search, and inference-time benchmarks are conventional code (Rust / PyTorch) with deterministic logs and JSON artifacts. AI did not invent experimental numbers.

Other tools. Rust/cargo for the procedural bench and meta binaries. PyTorch CUDA for overnight hybrid search. matplotlib for figures. MiKTeX pdflatex for PDF builds. GitHub for versioned papers and artifacts.

8 Broader impact and reproducibility

Broader impact. This work targets cycle-local wavetable seam repair for virtual-analog / software-instrument workflows. Misuse potential is low. The method does not enhance speech for surveillance, does not remove watermarks from protected media, and operates only on short synthetic periods. Training and search use procedural sine+cliff generators. No private studio stems are required for the reported tables. The main resource cost is GPU time on a consumer RTX 3090 (order of hours for the 5k-gate freeze, Table 6). Authors should not over-claim speech enhancement or production mastering quality from these metrics.

Reproducibility. Holdout seed 20,260,719. Overnight search seed 1,902,771,841. Primary code: <https://github.com/reeldemo/reelsynth> (scripts/overnight_gpu_rl_arch.py, scripts/bench_sota_matrix.py, scripts/metrics_snr_sdr.py). Paper companion: <https://github.com/reeldemo/denoise-opt-meta>. Dependencies for the CUDA benches: Python 3.12, PyTorch 2.6+CUDA 12.4, NumPy/matplotlib. Rust/cargo builds the separate sound_bench hardness probe. JSON artifacts under brand/artifacts/ (sota_matrix.json, canonical_eval_dataset/, overnight_history.jsonl) freeze the numbers in Section 5. PESQ/STOI are omitted for non-speech cycles (domain mismatch). MUSHRA was not run.

Competing interests. None to declare.

9 Limitations

- Residual uses a procedural ideal (no-cliff sibling). Perceptual MOS and listening panels are out of scope.

- Bake metrics leave formal listening tests and speech quality metrics to separate studies.
- **PESQ / STOI / MUSHRA deferred.** Primary cycles are non-speech wavetable tiles. Speech metrics would domain-mismatch sine+cliff families, so PESQ/STOI numbers are absent here (none invented on synthetic tiles). MUSHRA needs human listeners and was *not* run. A speech-proxy secondary study would need OA speech snippets labeled as secondary. That remains out of scope for this narrow-seam paper.
- The overnight CUDA runner scores Python make_batch sine+cliff draws. A separate Rust sound_bench export of 20 family tiles (byte-aligned generator) is a secondary transfer block. The Python generative SOTA matrix is the primary multi-family table.
- Per-trial architecture width is capped for overnight throughput. Full Demucs-diffusion-scale stacks were screened out.
- The 5k-gate freeze reports measured champion and overnight branch-best residuals. Isolated GA/PPO/GA+PPO/full re-runs use a short labeled budget (same search seed). Those short runs are a different protocol from the multi-hour freeze.
- Classical VA/BLEP anchors use author or proceedings PDFs (stilson1996, nam2009polyblep, esqueda2016blamp).

10 Outlook: follow-up publication

A self-contained follow-up can treat *which sounds fail* as the primary object of study: stratified residual heatmaps over the ten procedural families, cliff-size conditioning, and family-specialist or mixture-of-experts denoisers selected by the same RL-GA outer loop. That paper would cite the present mean- R hybrid protocol as the baseline and ask whether beating $R > 0.999$ is mainly an architecture problem or a coverage problem over hard families.

11 Conclusion

We framed wavetable wrap discontinuity as cycle-local seam artifact repair and specified DenoiseOpt, a hybrid GA-PPO(+PBT) meta-search scored by prolonged residual R . Related work separates used anchors (Noise2Noise / speech N2N, PBT, aging evolution, ERL, ENAS/PPO, Wave-U-Net/Conv-TasNet, MoE, BLIT/BLAMP, DDSP) from screened contrasts (MAML, DARTS, Demucs, DiffWave, full GAN loops). At the reported overnight freeze the champion reaches $R \approx 0.9909$ versus DualCosine ≈ 0.8166 . On the frozen canonical holdout a compact inference favorite reaches $R \approx 0.991$ versus DualCosine $R \approx 0.825$. Across twenty multi-family draws, mean $\Delta R \approx +0.162$ vs DualCosine (Wilcoxon $p \approx 9 \times 10^{-5}$). Fixed MLP/CNN-on- R baselines improve DualCosine but trail the meta favorite. Scope

remains wavetable seam restoration. Companion code lives at <https://github.com/reeldemo/reelsynth> and <https://github.com/reeldemo/denoise-opt-meta>.

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